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## Lecture 6: Randomized/Probabilistic Algorithms

### Example 1: Dataset Comparison

$$X = x_1 x_2 \dots x_n, \qquad x_i \in \{0,1\}$$

$$Y = y_1 y_2 \dots y_n, \qquad y_i \in \{0,1\}$$

Question: $X \stackrel{?}{=} Y$

### Classical Communication Protocol:

Comparing the two datasets bit by bit:

- Communication overhead: $O(n)$
- Comparison overhead: $O(n)$ => $n = 10^{16} bits\ (B) \approx 1136\ TB$

$$v(X) = value(X) = \sum_{1}^{n} x_i \, . \, 2^{n-i}$$

$$\Pi(k) = \{p \in \mathbb{P} \mid p \leq k\}, \ \pi(k) = |\Pi(k)|$$

### Randomized Equal algorithm:

$$Input: \quad \underset{A}{X = x_1 x_2 \dots x_n}, \quad \underset{B}{Y = y_1 y_2 \dots y_n}, \quad x_i y_i \in \{0,1\}$$

1. $A$ randomly chooses a prime number $p \in \Pi(n^2)$
2. $A$ calculates the "fingerprint of $X$ => $s$ = v (X) mod p.
   a. **Note:** X is considered as a binary number
3. $A$ sends s and p to $B$
4. $B$ calculates the "fingerprint of Y => t = v (Y) mod p
5. $B$ compares whether s = t? Are the two fingerprints the same?
   a. Yes: equal -> A,
   b. No: unequal -> A

### Comparison overhead:

The Equal algorithm is a randomized communication protocol for the data comparison example. It drastically reduces the comparison effort. Comparison overhead was previously $O(n)$, now the overhead is reduced to $5 \rightarrow O(1)$

**Communication overhead:**

$$0 \leq p, s \leq n^2$$
$$m \in \mathbb{N} \rightarrow \lceil \log m \rceil \; Bits$$
$$l(s,p) \leq 2 * \lceil \log n^2 \rceil \leq 4 * \lceil \log n \rceil$$

For $n = 10^{16}$;

$$\leq 4 * 16 * \lceil \log 10 \rceil$$
$$\leq 4 * 16 * 4$$
$$= 256 \text{ Bits}$$

Instead of $10^{16}$, we would only transmit 256 bits.

$$Prob_{equal}[A \mid B]$$

$$X = Y => t = s$$
$$X \neq Y <= t \neq s$$

$$Prob_{equal}[\text{"}unequal\text{"} \mid X = Y]) = 0$$

Example:

$$n = 5$$

$$X = 10011 \quad v(X) = 19$$
$$Y = 10001 \quad v(Y) = 17$$

$$for\; p = 11 => \text{ s } = \text{ 8, t } = \text{ 6}$$

$$X = 10011 \quad v(X) = 19$$
$$Y = 10001 \quad v(Y) = 17$$

$$for\; p = 2 => \text{ s } = \text{ 1, t } = \text{ 1}$$

$\underline{p = 2 \text{ is a bad witness for the equality of X and Y}}$

**Question:** How many bad witnesses are in $\Pi(n^2)$?

$s = t$, i.e. $v(x) = v(y) \; mod \; (p), although \; x \neq y.$

$$\Pi^{+}(n^2, X, Y) = \{p \in \mathbb{P} \mid v(X) \neq v(Y)(p), X \neq Y\}$$

$$\Pi^{-}(n^2, X, Y) = \{p \in \mathbb{P} \mid v(X) = v(Y)(p), X \neq Y\}$$

Establishing the relationship between bad witnesses and total witness candidates:

$$Prob_{equal}[X = Y \mid X \neq Y]) = \frac{|\Pi^{-}(n^2, X, Y)|}{|\Pi(n^2)|} = \frac{\mathbf{2\ln n}}{\boldsymbol{n}}$$

$$for\ n = 10^{16} \ \ => \frac{2\ln 10^{16}}{10^{16}} \approx 0.7 * 10^{-14}$$

Multiple rounds of algorithm to further reduce the error:

$$O(l * \log n)$$

$$l = 10 => 0.7 * 10^{-144}$$

$$L \subseteq \Sigma^*, w \in \Sigma^*: w \overset{?}{\in} L$$

$$D = \{(x, y) \in \{0,1\}^n x \{0,1\}^n \mid n \in \mathrm{N}_{0,}, x \neq y\}$$

Given $w \in \{0,1\}^n x\{0,1\}^n$

1. $Prob_{equal}["w \in D" \mid w \notin D] = 0$
2. $Prob_{equal}["w \in D" \mid w \in D] \geq 1 - \frac{2\ln n}{n} \ \ => RP(\varepsilon(n)),\ \ \varepsilon(n)\ is\ the\ error\ bound$
3. Overhead: $O(\log n)$

**Example 2: Triangle Graph**

$$\Delta Graph = \ \{\langle G\rangle \mid G\ undirected\ graph\ that\ contains\ at\ least\ one\ triangle\}$$

1. $T\ (tester)$ randomly chooses an edge $\{a, b\}$
2. $T$ randomly chooses a node $c \neq \{a, b\}$
3. Test: Do $c$ and $\{a, b\}$ form a triangle? -> Yes or No

$Prob_T["yes" \mid G \notin \Delta Graph] = 0$ ➔ no false positive statement

$$G = (V, E)$$

$$|V| = n, |E| = m$$

$$Prob_T["yes\ in\ one\ round" \mid G \in \Delta Graph] \geq \frac{3}{m} * \frac{1}{n-2}$$

$$Prob_T["no\ in\ one\ round" \mid G \in \Delta Graph] \leq 1 - \frac{3}{m(n-2)}$$

$$Prob_T["no\ in\ l\ rounds" \mid G \in \Delta Graph] \leq \left(1 - \frac{3}{m(n-2)}\right)^l$$

$$Prob_T["yes\ in\ l\ rounds" \mid G \in \Delta Graph] \geq 1 - \left(1 - \frac{3}{m(n-2)}\right)^l$$

$$\left(1 + \frac{1}{k}\right)^k \xrightarrow{k \to \infty} e = 2.7182 \ldots$$

More general formulation with x instead of the special case with 1:

$$\left(1 + \frac{x}{k}\right)^k \xrightarrow{k \to \infty} e^x$$

Place a minus before x:

$$\left(1 + \frac{-x}{k}\right)^k \xrightarrow{k \to \infty} e^{-x}$$

We want to apply the Euler sequence for our probabilistic $\Delta Graph$ algorithm:

$$k = l, x = yl \ \Rightarrow (1-y)^l \xrightarrow{k \to \infty} e^{-yl}$$

$$y = \frac{3}{m(n-2)}, l = \frac{m(n-2)}{3} \ \Rightarrow y * l = 1$$

$$\left(1 - \frac{3}{m(n-2)}\right)^{\frac{m(n-2)}{3}} \approx e^{-1} \approx \frac{1}{e} \approx \frac{1}{2.7} < \frac{1}{2}$$

RP

1. $Prob_T["yes" \mid G \notin \Delta Graph] = 0$
2. $Prob_T["yes" \mid G \in \Delta Graph] \geq \frac{1}{2}$
3. $polynomiel$

**Outline:**

$Dataset \ \in RP\left(\frac{2 \ln n}{n}\right)$

$\Delta Graph \in RP\left(\frac{1}{2}\right)$

**Complexity Class RP:**

Random/Probabilistic polynomial running algorithms with one-sided error

- No false positive statements.
- Conversely, RP algorithms make errors with a particular bound $\varepsilon$.
- By repeating the execution of the algorithm, the total error can be reduced.

⇨ The algorithms that class RP defines are also called **Monte Carlo algorithms**. Such algorithms allow a one-sided error.